Readers Are Functionals- Representation Control of the Relation Layer on Token Arenas

a multiplicity table, a faithful-reader certificate, and an isotypic correction to the rank budget

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Abstract

Attaching 2-cells to a token graph and computing invariant cohomology under a symmetry group yields a ledger of ranks, interactions, residues and containments. Papers I and II of this series reported such ledgers as facts about a substrate, or at best about a substrate-reader pair. We show that over $\mathbb{Q}$ with a finite reader group, the ledger is neither. Every quantity it reports at a reader H is $\dim V^H$ for a module V fixed independently of H, hence the inner product

$$\dim V^H = \langle m(V), f_H \rangle$$

of a reader-free multiplicity vector against the reader's fixed-point profile $f_H(\lambda) = \dim \lambda^H$. The relation field is recorded once; a reader only evaluates it.

The general layer that makes this work is classical and is used here with attribution rather than claimed: that rational representations are determined by their characters (Serre, Thm. 4 Cor. 2), that they are detected on cyclic subgroups (Artin, Serre Thm. 17), and the Burnside-ring analogue that G-sets are determined by fixed-point counts (Burnside; Bouc, Thm. 2.3.2). Assembled, these give the consequences we need. The recoverability loss satisfies $L_H(R|S) = \dim (Q_1(R|S))^H$, so loss is monotone along subgroup inclusion and the set of readers that see a containment is upward-closed: a containment cannot appear, vanish and reappear along a chain. Forced pairwise overlap must be counted inside each isotypic component before contraction, and the scalar rank budget understates it whenever the excess vector changes sign. Readers with $f_H > 0$ in every component — *faithful* readers — form a downward-closed ideal in the subgroup lattice, and a single equality at a faithful reader certifies a module equality at every reader. A nontrivial faithful reader exists exactly when some proper H has $\mathbb{Q}[G/H]$ containing every irreducible, which fails outright for G cyclic of prime order.

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On a 12-vertex C_4-bearing witness with $\mathrm{Aut}(X) \cong C_2 \times A_4$ — 26 subgroups in 12 conjugacy classes — we extract the full 9 × 6 multiplicity table from twelve measured reader columns. The fit is 54-fold overdetermined and solves exactly. The table then replaces the ledger. Residue zero at every reader is one module identity. The rotation-triangle family is the isotypic complement of the commuting squares plus one copy of the central sign character, which fixes both the saturation and the additivity pattern at a stroke. The spectator-square quotient is $3^+ \oplus 3^-$, whose subgroup shadow is the principal filter above the normal Klein four-group. The table predicts two forced overlaps that the scalar budget reports as zero; direct row-space intersection confirms them, 14 of 14. The minimal faithful apparatus on this witness uses 152 of 300 coordinates — strictly fewer than the raw field, and strictly more than the 22 of the full-automorphism reader, which is blind in five of six sectors and for that reason reported those two overlaps as independence.

We give the resulting method — inventory, deficiency, dominance shortlist, faithful-reader certification, propagation, evaluation — and show by measured counterexample that dominance is necessary and not sufficient: of four families dominating the commuting-square deficiency, two close it and two fall exactly one dimension short. All arithmetic is exact over $\mathbb{Q}$.

1. Introduction

A token arena is the state graph of K indistinguishable tokens moving on a finite simple graph X: vertices are the K-subsets of $V(X)$, adjacent when one token slides along a base edge into an unoccupied vertex. Attaching 2-cells along chosen closed walks and computing cohomology invariant under a group $G \leq \mathrm{Aut}(X)$ produces a first-cohomology residue $\varepsilon_{\mathrm{harm}}$, and comparing cell families produces a ledger: which family reads how much, which families overlap, which family is indispensable, which vocabulary closes the residue.

Papers I and II [1,2] built such ledgers and read them as structure of the substrate. That reading is wrong in a specific and correctable way. The same substrate read under two groups gives two different containment lattices: on the witness of this paper, spectator squares are contained in the span of spectator triangles under the full automorphism group and are not under the trivial group, by six dimensions. One might then retreat to the position that the ledger describes the pair (X, G). That retreat is also wrong, and in the more interesting direction.

Over $\mathbb{Q}$ a finite group has semisimple representation theory, and taking H-invariants is exact. Every family image, every intersection, every quotient appearing in the ledger is therefore a rational G-module, fixed before any reader is chosen, and the reader enters only by selecting fixed vectors. The ledger is a function of X alone, recorded as a table of multiplicities, and G is a point of evaluation. Nothing in the field is created or destroyed by the reader; what changes is which isotypic components have a fixed readout.

This has operational teeth, not only interpretive ones. Three errors follow from ignoring it, and all three were present in the earlier ledgers. A reader-invisible module gets called absent. Slack in a scalar rank budget gets called genuine independence. Multiplicity coverage gets called an actual span. Sections 4, 5 and 10 separate these into three distinct algebraic conditions, and Section 9 exhibits a case where the second error silently reports two families as exactly independent when the underlying modules are forced to share a dimension.

Sections 2–5 are general: the apparatus, the evaluation theorem, the isotypic capacity bound, and the faithful-reader lemma. Sections 6–10 are a single worked witness, chosen in Paper II as a counterexample to a different

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conjecture and reused here because its automorphism group is large enough to have a nontrivial subgroup lattice and small enough to enumerate exactly. Section 11 states the method the general theory licenses; Section 12 records scope, including two corrections to the framing of Papers I and II.

2. The apparatus

2.1 Arena, cells, invariant complex

Let X be a finite simple graph on n vertices and fix $1 \le K \le n$. The *arena* $A_K(X)$ has vertex set the K-subsets of $V(X)$, with $S \sim T$ when $|S \triangle T| = 2$ and the symmetric difference is an edge of X. Fix $G \le \mathrm{Aut}(X)$; it acts on the arena.

Let $C^1{}_G$ be the space of G-invariant rational 1-cochains on the arena: one coordinate per G-orbit of oriented arena edges, with an orbit containing an edge and its reverse forced to zero. Let d_0 be the coboundary from G-invariant 0-cochains, i.e. one row per vertex orbit, and set

$$\varepsilon \;=\; \dim C^1{}_G \;-\; \operatorname{rank} d_0 \,.$$

A *relation family* $\mathcal{K}$ is a set of closed arena walks (here: 3-cycles and 4-cycles). Each cell $c \in \mathcal{K}$ evaluates a 1-cochain by summing it with sign along c; this gives a row $d_1(c) \in (C^1{}_G)^*$, and because $d_1 d_0 = 0$ every such row annihilates $\operatorname{im} d_0$. Writing

$$A_G \;=\; (\, C^1{}_G \,/\, \operatorname{im} d_0 \,)^* \;, \quad \dim A_G = \varepsilon \,,$$

every family determines a subspace $W_F \le A_G$, the span of its cell rows, with $\operatorname{rank} d_1(\mathcal{K}) = \dim W_F$ and

$$\varepsilon_{\mathrm{harm}}(\mathcal{K}) \;=\; \dim \mathcal{H}^1{}_G(\mathcal{K}) \;=\; \varepsilon - \operatorname{rank} d_1(\mathcal{K}) \;=\; \dim A_G / W_{\mathcal{K}} \,.$$

So the residue is a quotient, and ε is a hard ceiling on any family's rank. We call A_G the *ambient module* and the W_F the *family submodules*. Both are G-stable.

2.2 The five families

A 4-cycle of the arena is one of three kinds, classified by the multiset of base edges its four moves use. If exactly two distinct base edges appear, each twice, the cycle is a *commuting square*: two tokens move along edges of disjoint closed support, in either order. Otherwise one token circulates a base 4-cycle while the others sit still; call it a *spectator square* if all four arena vertices share a common occupied vertex and a *rotation square* otherwise. Triangles split the same way into *spectator* and *rotation* 3-cycles. We write

$$\mathfrak{F} \;=\; \{\, \mathcal{K}_{\mathrm{disj}},\ \mathrm{spec}4,\ \mathrm{rot}4,\ \mathrm{spec}3,\ \mathrm{rot}3 \,\} \,,$$

and $\mathcal{K}_4 = \mathcal{K}_{\mathrm{disj}} \cup \mathrm{spec}4 \cup \mathrm{rot}4$ for all squares, $\mathcal{K}_{3,4}$ for the union of all five. Paper II showed $\mathcal{K}_4 = \mathcal{K}_{\mathrm{disj}}$ exactly when X has no C_4 subgraph [2]; the present paper works on a substrate where they differ, so all five families are populated.

2.3 The ledger quantities

For families R, S and reader $H \le G$, write $r_H(R) = \dim W^H{}_R$ and $D_H = \dim A^H$. The ledger reports

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$$I_H(R,S) = r_H(R) + r_H(S) - r_H(R \cup S)\,, \quad L_H(R|S) = r_H(R \cup S) - r_H(S)\,,$$

the *interaction* and the *loss*. By the dimension formula for subspace sums, I_H is the dimension of the intersection of the two row spaces, so

$$L_H(R|S) \;=\; r_H(R) - I_H(R,S) \;=\; \dim Q_H(R|S)\,, \quad Q_H(R|S) = W^H{}_R \,/\, (\, W^H{}_R \cap W^H{}_S\,)\,.$$

This is worth stating because it closes off a line of enquiry before it is opened. One naturally asks, of a redundant vocabulary, which other families determine a given one; for two families that question returns I_H read from the other side, and is not a second measurement. Only leave-one-out against three or more families carries information the pairwise table does not, because inclusion–exclusion fails there.

Finally, $R \preceq_H S$ means $W^H{}_R \le W^H{}_S$, equivalently $L_H(R|S) = 0$. The subscript is not decoration: Section 8 exhibits a pair with $\mathrm{spec}_4 \preceq \mathrm{spec}_3$ under one reader and not under another on the same substrate.

3. Reader evaluation

This section and the two that follow are standard representation theory, assembled in the form the application needs. Nothing in Sections 3–5 is claimed as new; the contribution of this paper begins at Section 6. We state the results with proofs because the exact hypotheses matter downstream and because two of them were, in earlier drafts of this work, used in incorrect forms.

Let $\mathrm{Irr}_{\mathbb{Q}}(G)$ be the rational irreducible representations of G, and for a finite-dimensional rational G-module V let $m(V) \in \mathbb{Z}_{\ge 0}{}^{\mathrm{Irr}}$ be its multiplicity vector. For $H \le G$ define the *fixed-point profile*

$$f_H(\lambda) \;=\; \dim \lambda^H \;=\; (1/|H|)\, \Sigma_{h \in H}\, \chi_\lambda(h)\,.$$

> **Theorem 1. (reader evaluation)** For every finite-dimensional rational G-module V and every $H \le G$, $\dim V^H = \langle\, m(V), f_H \,\rangle$.

Proof. $V \cong \bigoplus_\lambda \lambda^{\oplus m}$ with $m = m_\lambda(V)$, and taking H-invariants commutes with finite direct sums. ■

Theorem 1 is elementary. What is not elementary is the observation that every number in the ledger is of this form, which requires knowing that the modules in question do not depend on H. That is the content of the next result.

> **Theorem 2. (the quotient theorem)** Let W_R, $W_S \le A$ be the family submodules at the trivial reader and $Q_1(R|S) = W_R/(W_R \cap W_S)$. Then for every $H \le G$, $L_H(R|S) = \dim (\, Q_1(R|S)\,)^H$.

Proof. Over $\mathbb{Q}$ with $|H|$ invertible, restriction of functionals to $C^1{}_H \le C^1{}_1$ is the Reynolds projector onto H-invariants, so $W^H{}_F = (W_F)^H$ for every family F — both sides are spanned by the images of the same cells. Taking H-invariants is exact by Maschke, hence

$$Q_H = (W_R)^H \,/\, (W_R \cap W_S)^H = (\, W_R/(W_R \cap W_S)\,)^H = (Q_1)^H\,. \;\blacksquare$$

The same argument applies verbatim to W_R itself, to the intersection, to a union of families, and to the leave-one-out quotient, since each is spanned by cell images. So every ledger quantity at every reader is one inner product against a vector fixed once and for all. The practical consequence is that a subgroup walk is not a

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sequence of experiments: it is one representation sampled through different fixed-vector functionals, and once the representation is known the walk is redundant.

> **Corollary 2.1. (monotonicity)** If $H_1 \le H_2$ then $L_{H_2}(R|S) \le L_{H_1}(R|S)$. Hence Switch$(R,S) = \{ H : L_H(R|S) = 0 \}$ is upward-closed in the subgroup lattice.

Proof. $\lambda^{H_2} \le \lambda^{H_1}$ for every λ, so $f_{H_1} \ge f_{H_2}$ componentwise; apply Theorems 1 and 2. Alternatively and without characters: restriction induces a surjection $Q_{H_1} \twoheadrightarrow Q_{H_2}$. ■

Three regimes are conceivable a priori for a containment as the reader grows — upward-closed, non-monotone, or an antichain of minimal switches. Corollary 2.1 eliminates the second before any computation. A containment cannot appear, disappear and reappear along an inclusion chain. An antichain of minimal elements remains possible in general; on the witness of Section 8 it does not occur.

> **Proposition 3. (two collapses)** (i) The induced map $Q_1(R|S) \to Q_H(R|S)$ is surjective, so dim ker = $L_1(R|S) - L_H(R|S)$. (ii) $Q_H(R|S)$ has dimension $L_H(R|S)$.

Both are worth recording because both have been proposed as new invariants. They are not. The statement that a reader annihilates exactly the previously independent quotient is automatic whenever containment appears, by (i); and naming the loss as a quotient dimension, by (ii), says what the number is without adding a measurement. The *kernel as a subrepresentation* is content — it is a sub-object of Q_1 with an isotypic label — but its dimension is subtraction.

4. Capacity is vector-valued

Since rank d_1 cannot exceed D_H, two families whose ranks sum above D_H must overlap for reasons that have nothing to do with their shapes. Subtracting that forced part is the natural correction, and the obvious form of it is

$$I^{scal}_{cap}(R,S;H) \;=\; \max(\,0,\, r_H(R) + r_H(S) - D_H\,)\,.$$

This is too coarse, and in a way that inverts conclusions. Forced overlap occurs inside each isotypic component separately: a spare copy of one irreducible cannot absorb pressure in a different one.

> **Theorem 4. (isotypic capacity)** For submodules $R, S \le A$ and any $H \le G$, $I_H(R,S) \ge \Sigma_\lambda \max(\,0, m_\lambda(R) + m_\lambda(S) - m_\lambda(A)\,) \cdot \dim \lambda^H =: I^{iso}_{cap}(R,S;H)$.

Proof. Let $J = R \cap S$. Inside the λ-isotypic component $A_\lambda \cong \lambda^{\oplus m}$, submodules correspond to subspaces over the endomorphism field of λ, and the dimension formula in that field gives $m_\lambda(J) \ge \max(0, m_\lambda(R) + m_\lambda(S) - m_\lambda(A))$. Now apply Theorem 1 to J. ■

> **Proposition 4.1. (when the two caps agree)** Write $e_\lambda = m_\lambda(R) + m_\lambda(S) - m_\lambda(A)$. Then $I^{scal}_{cap} = \max(0, \Sigma_\lambda e_\lambda \dim \lambda^H)$ while $I^{iso}_{cap} = \Sigma_\lambda \max(0, e_\lambda) \dim \lambda^H$. The two agree if and only if the e_λ have one sign on the components λ with $\dim \lambda^H > 0$, and otherwise $I^{iso}_{cap} > I^{scal}_{cap}$.

In words: the scalar formula takes the maximum *after* the channels have been added together, so positive pressure in one irreducible can be cancelled on paper by unused capacity in another. That cancellation is an

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artifact of the collapse. The correction matters: Section 9 exhibits two family pairs for which the scalar cap is 0 at every reader while the isotypic cap is 1 at five of twelve, and direct computation confirms the overlap is 1.

One consequence deserves to be stated on its own, because it is the error the earlier ledgers made. A measured $I_H(R,S) = 0$ at some reader does *not* establish that R and S are independent as modules. It establishes that the reader has no fixed vector in their intersection. If that intersection is nonzero and lies in a component with $\dim \lambda^H = 0$, the overlap is real and the reader cannot see it.

5. Faithful readers

Multiplicity arithmetic records an inventory and not an embedding. If $m(U) = m(V)$ for $U \le V$ we may conclude $U = V$; but a multiplicity *identity* among sums, of the kind that appears when one asks whether two families jointly span the ambient, does not by itself force a span equality. Two multiplicity-one submodules of $\lambda^{\oplus 2}$ can coincide rather than span both copies. The following lemma is the certificate that closes that gap.

> **Lemma 5. (faithful-reader propagation)** Call $H \le G$ faithful if $\dim \lambda^H > 0$ for every $\lambda \in \mathrm{Irr}_{\mathbb{Q}}(G)$, equivalently if $\mathbb{Q}[G/H]$ contains every irreducible. Let $U \le V$ be rational G-modules and H faithful. If $\dim U^H = \dim V^H$ then $U = V$, and therefore $\dim U^K = \dim V^K$ for every $K \le G$.

Proof. $m(V) - m(U) \ge 0$ componentwise since $U \le V$. By Theorem 1, $\langle m(V) - m(U), f_H \rangle = 0$, and $f_H > 0$ componentwise, so the difference vanishes. Equal multiplicities with $U \le V$ and semisimplicity give $U = V$. ■

So one equality at one faithful reader propagates to the whole subgroup poset. A twelve-reader saturation table is not twelve pieces of evidence; it is one certificate and eleven redundant confirmations.

> **Proposition 6. (faithfulness is an order ideal)** If $H_1 \le H_2$ and H_2 is faithful then H_1 is faithful. Hence the faithful readers form a downward-closed subset of the subgroup lattice, and the smallest apparatus — least $\dim C^1_H$ — is attained at a maximal faithful subgroup.

Proof. $f_{H_1} \ge f_{H_2}$ componentwise, as in Corollary 2.1. ■

The naming inverts, and it is worth saying plainly: the *largest* faithful subgroup gives the *smallest* complete apparatus. Note also what Proposition 6 removes. Asking for "the cheapest reader that still sees everything" sounds like an optimization over an unspecified domain; it is not. The domain is the subgroup lattice, the admissible set is an ideal in it, and the answer sits on that ideal's frontier.

> **Proposition 7. (groups with no faithful proper reader)** If G is cyclic of prime order then the only faithful reader is the trivial one.

Proof. The only proper subgroup is trivial, and a nontrivial irreducible of C_p has no G-fixed vector. ■

So the existence of a nontrivial faithful reader — of any reduction of the apparatus that blinds no sector — is a property of the reader group, decidable from its character table, and not a property of token arenas. Any substrate with $\mathrm{Aut}(X) \cong C_p$ has no such reduction. This is the correct adversarial test for the results of Section 8, and it can be settled without running a substrate at all.

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5.1 A remark on complements

Maschke guarantees that every family submodule $W \le A$ has a G-stable complement W' with $A = W \oplus W'$. The isotypic decomposition $A = \bigoplus_\lambda A_\lambda$ is canonical; the complement is not. A submodule of $\lambda^{\oplus m}$ with $m > 1$ admits many G-stable complements, which is the same ambiguity Lemma 5 exists to resolve, and it should not be elided.

What is determined is the isomorphism class, hence the multiplicity vector: $W' \cong A/W$. And the direction of the statement matters. A transverse quotient exists when the vocabulary is *incomplete*:

$$\varepsilon_{\text{harm}} > 0 \iff W \subsetneq A \iff A/W \neq 0 .$$

A nonzero harmonic residue is precisely the nonzero quotient left by an incompletely spanning vocabulary. No metric and no notion of orthogonality is involved.

6. The witness

Fix X on 12 vertices with the 30 edges

01 02 04 05 06 07 09 12 13 15 17 18 1·10 23 24 28 29 2·11 34 35 36 39 3·10 45 47 4·10 4·11 56 58 5·11

Degree sequence $7^6 3^6$. The six vertices of degree 7 induce the octahedron $K_{2,2,2}$ with antipodal pairs {0,3}, {1,4}, {2,5}; each of the six vertices of degree 3 attaches to one full antipodal pair together with one vertex of the next pair cyclically. Girth is 3, with 20 triangles, and there are 66 four-cycle subgraphs of which 24 are chordless — the substrate is deliberately not C_4-free, so by Paper II [2] all three square families are populated. Throughout, $K = 2$.

The arena has 66 vertices and 300 edges. Direct enumeration gives $|\mathrm{Aut}(X)| = 24$ with 26 subgroups in 12 conjugacy classes, of orders

1 : 1, 2 : 7, 3 : 4, 4 : 7, 6 : 4, 8 : 1, 12 : 1, 24 : 1,

which identifies $\mathrm{Aut}(X) \cong C_2 \times A_4$: a central involution z, the unique index-2 subgroup A_4, and the normal Klein four-group $V_4 \lhd A_4$. Its six rational irreducibles, on the rational classes $(1, z, v, zv, t, zt)$ of sizes (1, 1, 3, 3, 8, 8) — where $v \in V_4 \setminus 1$, t has order 3, and rational classes fuse g with g^{-1} — have characters

λ	dim	1	z	v	zv	t	zt
1^+	1	1	1	1	1	1	1
1^-	1	1	−1	1	−1	1	−1
2^+	2	2	2	2	2	−1	−1
2^-	2	2	−2	2	−2	−1	1
3^+	3	3	3	−1	−1	0	0
3^-	3	3	−3	−1	1	0	0

Table 1. Rational character table of Aut(X) ≅ $C_2 \times A_4$. The 3± are the standard 3-dimensional A_4-representation tensored with the two characters of the central C_2; 2± have endomorphism field $\mathbb{Q}(\omega)$.

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The resulting fixed-point profiles f_H are the reader side of every measurement in this paper, and nothing else about the reader enters.

id	|H|	class	z ∈ H	H ≤ A_4	1^+	1^-	2^+	2^-	3^+	3^-	faithful
0	1	1	no	yes	1	1	2	2	3	3	yes
1	2	3	no	no	1	0	2	0	1	2	—
2	2	3	no	yes	1	1	2	2	1	1	yes
3	2	1	yes	no	1	0	2	0	3	0	—
4	3	4	no	yes	1	1	0	0	1	1	—
5	4	3	no	no	1	0	2	0	0	1	—
6	4	3	yes	no	1	0	2	0	1	0	—
7	4	1	no	yes	1	1	2	2	0	0	—
8	6	4	yes	no	1	0	0	0	1	0	—
9	8	1	yes	no	1	0	2	0	0	0	—
10	12	1	no	yes	1	1	0	0	0	0	—
11	24	1	yes	no	1	0	0	0	0	0	—

Table 2. The twelve conjugacy classes of subgroups of Aut(X) and their fixed-point profiles. id 7 is V_4, id 9 is $C_2 \times V_4$, id 10 is A_4, id 11 the whole group. Exactly two classes are faithful.

6.1 The measured ledger

One representative per class, exact rational elimination throughout. Cell counts are reader-independent: 291 commuting squares, 528 spectator squares, 264 rotation squares, 180 spectator triangles, 20 rotation triangles, 1283 in all.

id	|H|	dim C^1	rank d_0	D	disj	spec4	rot4	spec3	rot3	r(ALL)	resid	L(s4|s3)	uniq disj
0	1	300	65	235	216	168	150	174	20	235	0	6	12
1	2	186	47	139	127	92	84	99	12	139	0	3	8
2	2	152	37	115	106	80	70	86	10	115	0	2	6
3	2	150	35	115	105	84	72	87	10	115	0	3	6
4	3	100	21	79	72	56	50	58	8	79	0	2	4
5	4	112	33	79	72	48	44	55	7	79	0	1	5
6	4	94	27	67	61	44	38	49	6	67	0	1	4
7	4	78	23	55	51	36	30	42	5	55	0	0	3
8	6	50	11	39	35	28	24	29	4	39	0	1	2
9	8	66	23	43	39	24	21	30	4	43	0	0	3
10	12	26	7	19	17	12	10	14	3	19	0	0	1
11	24	22	7	15	13	8	7	10	2	15	0	0	1

Table 3. The twelve-reader ledger. D = dim C^1_H – rank d_0. Family columns are rank d_1. 'resid' is D – r(ALL); 'uniq disj' is the leave-one-out loss L($\mathcal{K}$_disj | the other four).

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Four features of Table 3 hold at every reader, and we will see in Section 8 that each is one module fact rather than twelve. The residue of the full vocabulary vanishes throughout, from $D = 15$ to $D = 235$. $\mathcal{K}_{\text{disj}}$ is the only family with nonzero leave-one-out loss — every other family is fully recoverable from the remaining four under every reader — although its loss ranges from 1 to 12, so the *location* of the irredundant channel is reader-invariant while its dimension is not. Among all 20 ordered family pairs at all 12 readers, spec4 $\preceq$ spec3 is the only containment that ever occurs. And the minimal saturating vocabularies are $\{\mathcal{K}_{\text{disj}}, \text{spec3}\}$ and $\{\mathcal{K}_{\text{disj}}, \text{rot3}\}$ at every reader, out of 31 nonempty subfamilies.

Monotonicity of Corollary 2.1 was checked directly across the poset: 0 violations over 1000 ordered (family-pair, inclusion) tests.

At the full-automorphism reader the leave-one-out line admits an explicit witness rather than a rank count. With $\mathcal{K}_{\text{disj}}$ removed, $\ker d_1$ has dimension 8 over a rank-7 image of d_0, so $\mathcal{H}^1$ is one-dimensional, with integral representative supported on 3 of the 22 invariant coordinates. Of the 1283 cells, exactly 96 detect it — all commuting squares, using 96 distinct disjoint base-edge pairs, one cell each — and 0 of 528 spectator squares, 0 of 264 rotation squares, 0 of 180 spectator triangles and 0 of 20 rotation triangles detect it.

7. The multiplicity table

By Theorems 1 and 2, each column of Table 3 is $\dim V^H$ for one reader-free module, so it must satisfy $\langle m(V), f_H \rangle$ = measured value simultaneously for all twelve H. That is twelve equations in six unknown multiplicities. Solving is therefore also a consistency test, 54-fold overdetermined across the nine columns we extract. *All nine solve exactly.* No arena is rebuilt and no rank is recomputed in this section; the input is Table 2 and Table 3.

module	1^+	1^-	2^+	2^-	3^+	3^-	dim
C^1 (trivial reader)	22	4	22	4	28	46	300
ambient A = (C^1/im d_0)*	15	4	14	4	24	36	235
W($\mathcal{K}$_disj)	13	4	13	4	22	33	216
W(spec4)	8	4	8	4	20	24	168
W(rot4)	7	3	7	3	17	23	150
W(spec3)	10	4	10	4	19	25	174
W(rot3)	2	1	1	0	2	3	20
Q(spec4 | spec3)	0	0	0	0	1	1	6
the irredundant $\mathcal{K}$_disj channel	1	0	1	0	1	2	12

Table 4. The 9 × 6 multiplicity table. Every number in Table 3, and every interaction, loss, residue and containment derived from it, is an inner product of a row of this table with a row of Table 2.

The solution is unique, and this needs checking rather than assuming: a consistent overdetermined system still has a one-parameter family of solutions if its coefficient matrix is rank-deficient. The twelve profiles of Table 2, as a 12 × 6 integer matrix, have rank 6 with zero nullspace, so $m(V)$ is determined. Moreover the six *cyclic* classes — orders 1, 2, 2, 2, 3, 6 — already have rank 6 on their own, and are the smallest subset that does.

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That is the expected shape rather than an accident. Over a field of characteristic zero a representation is determined by its character [6, Thm. 4 Cor. 2]; a rational character is detected on cyclic subgroups by Artin's induction theorem [6, Thm. 17]; and dim $V^H = (1/|H|)\,\Sigma_{h\in H}\,\chi(h)$, so Möbius inversion over the divisor lattice of each cyclic subgroup recovers χ from the fixed-point dimensions. The exact analogue for G-sets is Burnside's theorem on marks — two G-sets are isomorphic iff $|X^H| = |Y^H|$ for every H — and the injectivity of the mark homomorphism [8, Thm. 2.3.2 and §3.2]. The extraction of Table 4 is an instance of that classical detection principle, not a new one, and the rank computation above is its verification in this case.

Three columns recovered this way, against their measured values, by way of illustration: $r_H(\text{rot3}) = 20, 12, 10, 10, 8, 7, 6, 5, 4, 4, 3, 2$; $L_H(\text{spec4}|\text{spec3}) = 6, 3, 2, 3, 2, 1, 1, 0, 1, 0, 0, 0$; and the leave-one-out loss of $\mathcal{K}_{\text{disj}} = 12, 8, 6, 6, 4, 5, 4, 3, 2, 3, 1, 1$.

The scope of the resulting statement should be stated exactly. Within the finite-reader rational representation apparatus, the relation field is completely represented, *for reader evaluation*, by its multiplicity vector. The reader does not change the multiplicity module; it changes only which fixed-vector dimensions are visible. This is not a tensor factorization of the relation space, and it is not a claim about fields in any other sense.

8. What the table says

8.1 One line

Define the *deficiency* of a family as what it cannot reach, as a module: $\delta(F) = m(A) - m(W_F)$. Then Table 4 gives

$$m(\text{rot3}) \;=\; \delta(\mathcal{K}_{\text{disj}}) \;+\; (0, 1, 0, 0, 0, 0)\,, \quad \delta(\mathcal{K}_{\text{disj}}) = (2, 0, 1, 0, 2, 3)\,.$$

The rotation-triangle family is the exact isotypic complement of the commuting squares in the ambient module, plus one redundant copy of the central sign character 1^-. Six integers. Together with a single span certificate at a faithful reader (Section 10), this one line accounts for:

— $r_H(\mathcal{K}_{\text{disj}} \cup \text{rot3}) = D_H$ at all twelve readers, so $W_{\text{ALL}} = A$ and the residue vanishes identically rather than coincidentally;

— $l_H(\mathcal{K}_{\text{disj}}, \text{rot3}) = \dim (1^-)^H$, hence 1 exactly when $H \le A_4$ and 0 otherwise, which is the whole additivity pattern across Table 3;

— $\varepsilon_{\text{harm}}(\mathcal{K}_{\text{disj}}) = 19$ at the trivial reader and 2 at $\text{Aut}(X)$, as two contractions of the same deficiency vector;

— the rank-access result of Paper I §4, that rotation triangles read the dimension-2 residue at full rank while all 1083 squares read one line [1].

The shared line admits a direct description. At the trivial reader $W(\mathcal{K}_{\text{disj}}) \cap W(\text{rot3})$ is one-dimensional, spanned by an integral cochain functional of support 24 of 300 with entries in $\{-1, 0, +1\}$; both families are stable under every automorphism, so the line is too, and the central involution acts on it by

$$z \cdot v \;=\; -\,v\,.$$

Restricting v to invariant coordinates leaves 2 nonzero coordinates under A_4, and 0 under $\langle z\rangle$ and under $\text{Aut}(X)$. So the line is the sign character of $\text{Aut}(X)/A_4$, and Reynolds averaging over any reader meeting the non-A_4

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coset cancels it. One mechanism was eliminated on the way: a reversible arena-edge orbit is forced to zero in the invariant basis and would delete a coordinate, but the witness arena has *o* reversible orbits at every one of the twelve readers — edge-orbit count equals dim $C^1{}_H$ exactly at all twelve — so that machinery never fires here.

8.2 The spectator-square quotient, and why V_4

Table 4 gives $Q_1(\text{spec}4|\text{spec}3) = 3^+ \oplus 3^-$, of dimension 6 with $\langle\chi,\chi\rangle = 2$ forcing two distinct irreducibles each of multiplicity one, which in $C_2 \times A_4$ can only be 3 + 3. By Theorem 2 this determines the entire containment behaviour:

$$L_H(\text{spec}4|\text{spec}3) = \dim (3^+)^H + \dim (3^-)^H ,$$

and reading Table 2, both vanish exactly at ids 7, 9, 10, 11 — V_4, $C_2 \times V_4$, A_4 and the whole group. Hence

$$\text{Switch}(\text{spec}4, \text{spec}3) = \uparrow V_4 ,$$

the principal upward filter generated by the normal Klein four-group, with a unique minimal element rather than an antichain. And the reason is representation-theoretic, not coincidental: the standard 3-dimensional A_4-representation restricted to V_4 is the sum of V_4's three nontrivial characters and contains no trivial summand, so V_4 is the smallest reader with no fixed vector in the quotient.

The six spectator-square directions do not go anywhere at V_4. They carry an isotypic label, and a reader sees them exactly while its restriction of that label still has invariants. This is the precise distinction the earlier ledgers lacked:

condition	diagnosis
m_λ(V) = 0	genuine absence — the channel is not in the module
m_λ(V) > 0 and dim λ^H = 0	reader-induced flatness — the channel exists, this reader has no fixed readout of it
m_λ(F) < m_λ(V)	vocabulary deficiency — the family does not reach the available multiplicity in that sector

Table 5. Three algebraically distinct conditions that an unexamined ledger reports as the same zero. A zero at a reader is an evaluation result; a zero in the multiplicity vector is an existence result.

8.3 The minimal faithful apparatus

By Proposition 6 the faithful readers form an ideal; reading the profile columns of Table 2, exactly two classes are faithful, and the ideal has a unique maximal element.

reader	dim C¹_H	D_H	f_H	what it is
trivial	300	235	(1,1,2,2,3,3)	raw field; every channel fully visible
order 2 inside A_4	152	115	(1,1,2,2,1,1)	minimal faithful apparatus; every channel still visible
Aut(X)	22	15	(1,0,0,0,0,0)	maximal compression; only the trivial sector visible

Table 6. Three readers, three different things. The middle row is the maximal faithful subgroup class, hence the smallest complete apparatus.

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The order-2 class inside A_4 is faithful because its generator lies in A_4, so 1^- stays visible; V_4 lies in the kernel of A_4's linear characters, so $2^\pm$ stay full; and the standard 3-dimensional representation restricted to an involution retains a one-dimensional fixed space, so $3^\pm$ survive. Half the coordinates of the raw field, and no sector goes dark. Above the frontier sectors begin to vanish; below it the additional coordinates carry no new isotypic channel.

The third row explains the failure analysed next. The full-automorphism reader is extremely legible and blind in five of six sectors, so its zeros cannot be read as structural absence without Table 4. By Proposition 7 this hierarchy is not generic: a substrate with cyclic prime automorphism group admits no faithful reduction at all, and the 152 is a fact about $C_2 \times A_4$, not about token arenas.

9. The scalar budget fails, and the regression

Table 7 lists the excess vectors $e_\lambda = m_\lambda(R) + m_\lambda(S) - m_\lambda(A)$ for all ten family pairs, read off Table 4. By Proposition 4.1 the scalar and isotypic caps agree except where the excess changes sign on a visible component.

pair	1^+	1^-	2^+	2^-	3^+	3^-	mixed	readers where the caps differ
disj / spec4	6	4	7	4	18	21	no	—
disj / rot4	5	3	6	3	15	20	no	—
disj / spec3	8	4	9	4	17	22	no	—
disj / rot3	0	1	0	0	0	0	no	—
spec4 / rot4	0	3	1	3	13	11	no	—
spec4 / spec3	3	4	4	4	15	13	no	—
spec4 / rot3	−5	1	−5	0	−2	−9	yes	ids 0, 2, 4, 7, 10
rot4 / spec3	2	3	3	3	12	12	no	—
rot4 / rot3	−6	0	−6	−1	−5	−10	no	—
spec3 / rot3	−3	1	−3	0	−3	−8	yes	ids 0, 2, 4, 7, 10

Table 7. Excess vectors. Two pairs are mixed-sign, and in both the single positive entry is in the 1^- sector — the one Aut(X) cannot see.

The five readers named in the last column are exactly those with $H \le A_4$, i.e. with $\dim (1^-)^H = 1$. For spec4 / rot3 and spec3 / rot3 the table therefore predicts a forced overlap of 1 at those five readers, while the scalar cap reports 0 at all twelve. Every other family carries the ambient's full 1^- multiplicity 4, and rot3's single copy sits inside all of them; rot4 is the only family deficient in the minus sectors, with $m_{1^-} = m_{2^-} = 3$ against 4, which is exactly why rot4 / rot3 is the one rotation-triangle pairing with no forced overlap anywhere.

This prediction comes from character arithmetic on Table 4. We tested it against a direct row-space intersection, recomputed from the arena at each reader, which exercises the extraction pipeline independently of the arithmetic that produced it. Five readers inside A_4 and two controls outside:

id	|H|	$H \le A_4$	D_H	r(spec4)	r(spec3)	r(rot3)	I(spec4,rot3)	I(spec3,rot3)	iso cap	scalar cap
0	1	yes	235	168	174	20	1	1	1	0
2	2	yes	115	80	86	10	1	1	1	0

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id	|H|	H ≤ A_4	D_H	r(spec4)	r(spec3)	r(rot3)	I(spec4,rot3)	I(spec3,rot3)	iso cap	scalar cap
4	3	yes	79	56	58	8	1	1	1	0
7	4	yes	55	36	42	5	1	1	1	0
10	12	yes	19	12	14	3	1	1	1	0
3	2	no	115	84	87	10	0	0	0	0
9	8	no	43	24	30	4	0	0	0	0

Table 8. Direct row-space intersections. 14 of 14 agree with the isotypic cap and disagree with the scalar one; the isotypic bound of Theorem 4 is attained exactly.

The methodological statement is worth isolating. At the full-automorphism reader one measures I(spec4, rot3) = 0 and would, on the scalar account, record exact independence with no counting pressure. The underlying modules share a dimension. What the measurement established was that dim $(1^{-})^{\mathrm{Aut}} = 0$.

10. Dominance is necessary and not sufficient

Only a family whose multiplicity vector dominates $\delta(F)$ componentwise can close F's residue, since otherwise some isotypic channel is unreachable. Dominance is therefore a sound shortlist. It is not a closure theorem, for the reason given in Section 5: multiplicities record an inventory, not which copy of an irreducible a submodule occupies. The witness contains both outcomes, which settles the question by example rather than by caution.

Deficiencies, from Table 4:

family F	δ(F) = m(A) − m(W_F)	dim	families whose vector dominates δ(F)
$\mathcal{K}$ disj	(2, 0, 1, 0, 2, 3)	19	spec4, rot4, spec3, rot3
spec4	(7, 0, 6, 0, 4, 12)	67	disj, rot4, spec3
rot4	(8, 1, 7, 1, 7, 13)	85	disj, spec4, spec3
spec3	(5, 0, 4, 0, 5, 11)	61	disj, spec4, rot4
rot3	(13, 3, 13, 4, 22, 33)	215	disj

Table 9. Deficiency vectors and their componentwise dominators.

All four other families dominate $\delta(\mathcal{K}_{\mathrm{disj}})$. Certified at the maximal faithful reader of Table 6, where $D = 115$ and Lemma 5 makes one equality sufficient for all readers:

family F	dominates δ($\mathcal{K}$_disj)	r($\mathcal{K}$_disj ∪ F)	D	closes
spec4	yes	114	115	no
rot4	yes	114	115	no
spec3	yes	115	115	yes
rot3	yes	115	115	yes

Table 10. Dominance does not imply closure. Two of the four dominators fall exactly one dimension short.

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So Table 9 shortlists four candidates and Table 10 certifies two. The certificate is what licenses the claims of Section 8.1: $m(\mathrm{rot3}) = \delta(\mathcal{K}_{\mathrm{disj}}) + 1^{-}$ supplies the inventory, and the span equality at a faithful reader supplies the closure, which Lemma 5 then propagates to the whole poset. Neither step alone suffices, and an earlier draft of this work asserted the propagation from the inventory alone, which is invalid.

11. The method

The general results of Sections 3–5 license a procedure in place of the enumeration that produced Table 3. The ordering is not optional; each step supplies a precondition for the next.

1. **Inventory.** Extract $m(A)$ and $m(W_F)$ for the candidate families, either directly or — as here — by solving an overdetermined system from a few reader columns.
2. **Deficiency.** Compute $\delta(F) = m(A) - m(W_F)$ for the declared vocabulary.
3. **Shortlist.** Keep only families whose multiplicity vector dominates δ componentwise.
4. **Certify.** Test the span equality once, at a faithful reader — a maximal element of the ideal of Proposition 6.
5. **Propagate.** Lemma 5 extends that equality to every reader.
6. **Evaluate.** Use f_H to report any rank, loss, interaction or residue at any reader.

On the witness this replaces a search over 31 nonempty subfamilies crossed with 12 reader classes by a six-component subtraction, a four-way dominance test, and one rank computation. The remaining search is where it belongs: locating candidate families in the arena. Once their modules are known, reader enumeration is no longer a discovery mechanism.

Two cautions are part of the method rather than appended to it. Before concluding from $I_H = 0$ that two families are independent, compute the isotypic cap (Theorem 4). Before concluding from $\varepsilon_{\mathrm{harm}} = 0$ at one reader that a vocabulary closes, check that the reader is faithful (Lemma 5), and before concluding from a multiplicity identity that it closes, obtain the span certificate (Section 10).

A final remark on vocabulary. Cell families are methods in a precise sense: each family determines a submodule of the ambient relation module, and that submodule's multiplicity vector records its isotypic reach. The levels should be kept distinct — an individual cell is one relation row; a family is a selected set of cells; the method image is the submodule they span; the reach is its multiplicity vector; certified closure is a proven equality of spans. Commuting squares, spectator triangles and rotation triangles are three rows of Table 4, not three descriptions of one object.

12. Scope

12.1 What is standard, and what is new

The general layer is classical and is used here under attribution. Semisimplicity of rational representations of a finite group and exactness of H-invariants are Maschke [6]. Theorem 1 is a direct consequence of additivity

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of invariants. The detection principle underlying the extraction of Section 7 — that a rational representation is determined by its fixed-point dimensions, with cyclic subgroups already sufficient — follows from [6, Thm. 4 Cor. 2] and Artin's induction theorem [6, Thm. 17], and is the representation-theoretic counterpart of Burnside's theorem on marks [8, Thm. 2.3.2]. Corollary 2.1 and Proposition 6 are the elementary observation $\lambda^{H_2} \le \lambda^{H_1}$. The bound of Theorem 4 is the dimension formula for subspace sums applied inside each isotypic block. Lemma 5 is a formal consequence of Theorem 1 together with nonnegativity of multiplicity differences; we have not found it stated under a name, and we do not claim it as new. Proposition 7 is immediate. We state these with proofs because the exact hypotheses are load-bearing downstream, not to claim them.

What this paper contributes is the identification and the measurement: that the relation layer of a token arena is such a system at all, so that a ledger indexed by readers collapses to a multiplicity table indexed by irreducibles; the 9 × 6 table of this witness and the structural facts read off it; the demonstration that the scalar rank budget misdiagnoses forced overlap, with a measured instance where it reports independence and the modules share a dimension; the counterexample separating multiplicity dominance from span closure; and the resulting method of Section 11. The corrections of Section 12.3 to the earlier papers of this series are also part of the contribution.

One terminological point is left open. We call a reader *faithful* when every rational irreducible has a nonzero fixed vector, equivalently when $\mathbb{Q}[G/H]$ contains every irreducible constituent, equivalently $\langle 1_H{}^G, \chi \rangle > 0$ for every χ. A search of the standard references and of the permutation-character literature did not turn up an established name for this condition; if one exists, the terminology here should yield to it. Note that "faithful" is already used for a different property of a representation, and the clash is regrettable but local to this paper.

12.2 What is witness-specific

$\mathrm{Aut}(X) \cong C_2 \times A_4$; the 9 × 6 table and everything read off it — the complement identity for rot3, the 1^- overlap line and its z-anti-invariance, $Q_1(\mathrm{spec4}|\mathrm{spec3}) = 3^+ \oplus 3^-$ and the filter above V_4, $W_{\mathrm{ALL}} = A$, the minus-sector deficiency of rot4, the 152-coordinate minimal faithful apparatus, and the dominance failures of spec4 and rot4. None of these is claimed for another substrate. By Proposition 7 the last is known not to generalize.

12.3 Corrections to Papers I and II

Two framings in the earlier papers of this series should be read as superseded rather than silently amended. First, the scalar rank budget: ε remains a correct ceiling on total visible rank, and the budget-corrected interaction term built from it remains a genuine improvement over the raw interaction, but it is not a sound diagnostic of forced pairwise overlap. Theorem 4 and Table 8 give the correct form and an instance of the failure. Where Paper I or an intermediate record reports a pair as structurally independent on the strength of a vanishing scalar cap, that reading is withdrawn; the numbers stand, the attribution does not.

Second, reader-relativity. The observation that a containment lattice depends on the reader and not only on the substrate is correct and was the right first corrective, but it is weaker than the truth. The lattice depends on the substrate alone, recorded as a multiplicity table; the reader supplies an evaluation. Statements of the form "this is a property of the pair (X, G)" should be replaced by the module identity plus the reader functional that produces the reported number.

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The claim that a harmonic residue is in general the shape of a missing cell family, with its dimension that family's rank, is *not* established and is not asserted here. On this witness the residue of $\mathcal{K}_{\text{disj}}$ is matched by one available family (Section 8.1, Table 10), and that is a witness theorem. A residue may in general require a combination of families, a family whose image is too large or too small, families absent from the arena, or directions no catalogued family reaches. Which residues admit a family-level complement is open.

12.4 Reproducibility

All linear algebra is exact over $\mathbb{Q}$ using fraction arithmetic; no floating point enters at any stage. Ranks are computed by fraction-free Gaussian elimination with full pivoting over the ordered coordinate basis. The invariant cochain basis is built from G-orbits of oriented arena edges with reversible orbits forced to zero, and d_0 from vertex orbits; two independent implementations of the invariant complex — an orbit construction and a Reynolds projector — were used throughout the series and agree on all reported values. Subgroups were enumerated by closure under composition from all cyclic subgroups followed by iterated joins, giving 26 subgroups, then partitioned into 12 conjugacy classes by conjugation. Multiplicity extraction solves the twelve-equation, six-unknown system by exact row reduction and rejects any column whose augmented system is inconsistent; all nine reported columns are consistent, and the solution is verified against all twelve readers after solving.

Acknowledgement of method

Several claims in this paper were stated, run, and withdrawn before reaching the form above. A chord mechanism for the spectator-square containment — a chord in a base C_4 splitting a one-token circulation into two triangles — was eliminated by the witness itself, which has 24 chordless C_4s of 66 and exhibits the containment anyway. A reversible-orbit mechanism for the 1^- line was eliminated by direct count: zero reversible orbits at all twelve readers. The propagation of saturation from a multiplicity identity alone was invalid and is replaced by Lemma 5. Each of these is recorded because a killed claim is a constraint, and because the surviving statements are easier to trust alongside the discarded ones.

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